

Kelompok 7
Jaringan Terintegrasi PT. Digital Maju (Soal Kelompok 5)

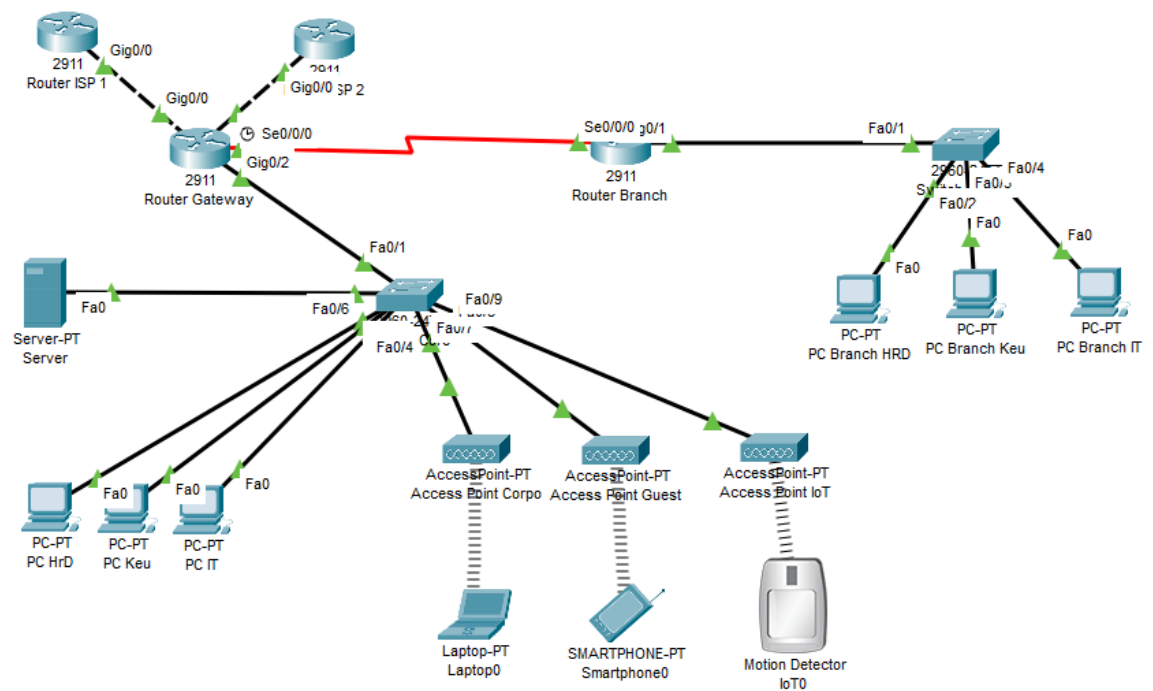
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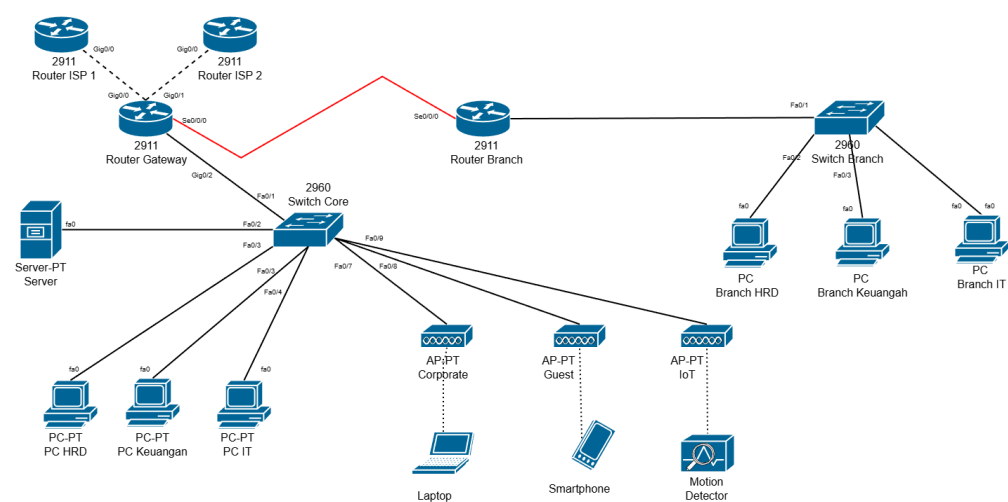
Jawaban:

1. Topologi Jaringan

1.1 Rancangan Topologi pada Cisco Packet Tracer



1.2 Diagram Topologi pada draw.io



1.3 Fungsi Utama Perangkat

Perangkat	Fungsi Utama dalam Skenario
Router-Gateway (HQ)	Gerbang Utama & Routing: Menghubungkan jaringan internal kantor pusat ke dunia luar (Internet) dan ke Kantor Cabang. Router ini bertanggung jawab atas NAT (menerjemahkan IP privat ke publik), Failover ISP (mengatur jalur utama/cadangan), dan Routing Antar-VLAN (Router-on-a-Stick).
Router ISP1 & ISP2	Penyedia Layanan Internet: Mensimulasikan koneksi ke dua provider internet berbeda untuk menjamin redundansi (ketersediaan tinggi). Jika ISP1 mati, ISP2 menggantikan.
Switch-Core	Distribusi & Segmentasi: Pusat pertemuan semua kabel dari departemen. Fungsinya memisahkan trafik berdasarkan VLAN (HRD, Keuangan, dll) agar tidak tercampur, serta menjaga keamanan dengan Port Security.
Server Farm	Penyedia Layanan Terpusat: Berisi server yang menjalankan: 1. DHCP: Memberikan IP otomatis ke PC/Laptop. 2. DNS: Menerjemahkan pt-digitalmaju.local menjadi IP address.
Access Point (AP)	Konektivitas Nirkabel: Menyediakan akses WiFi. AP ini harus mendukung Multi-SSID untuk memisahkan jaringan Karyawan (Corporate) dan perangkat pintar (IoT) secara logis.
Router-Branch	Penghubung Cabang: Menghubungkan staf di Bandung agar bisa berkomunikasi dengan server pusat di Jakarta secara aman.

2. Perencanaan IP Address & VLAN

2.1 Tabel alokasi IP Address dan VLAN

Departemen	VLAN/Interface	IP Network	Gateway	Rentang IP
ISP 1	Gi0/0	10.10.10.1	10.10.10.1	-
ISP 2	Gi0/0	20.20.20.1	20.20.20.1	-
HRD	10	192.168.10.0/24	192.168.10.1	192.168.10.2 - 192.168.10.254
Keuangan	20	192.168.20.0/24	192.168.20.1	192.168.20.2 - 192.168.20.254
IT	30	192.168.30.0/24	192.168.30.1	192.168.30.2 - 192.168.30.254
Manajemen/farm	40	192.168.40.0/24	192.168.40.1	192.168.40.2 - 192.168.40.254

3. Implementasi Layanan Jaringan

3.1 VLAN & Switching

Konfigurasi pada Switch-Core menggunakan logika:

1. Trunk Port: Port yang menuju ke Router Gateway dan Access Point dikonfigurasi mode Trunk (switchport mode trunk) untuk melewatkan trafik VLAN 10-60.
2. Access Port: Port yang menuju PC dikonfigurasi mode Access (switchport mode access) sesuai ID VLAN departemen.
3. VLAN Database: Seluruh VLAN ID didaftarkan di Switch Core agar trafik dapat dikenali.

3.2 Routing Antar VLAN

Untuk menghubungkan departemen yang berbeda subnet, digunakan teknik Router-on-a-Stick pada Router Gateway :

1. Interface fisik Gi0/1 diaktifkan tanpa IP Address.
2. Dibuat Sub-interface (contoh: Gi0/1.10) untuk setiap VLAN.
3. Enkapsulasi dot1Q diterapkan pada setiap sub-interface (encapsulation dot1q 10).
4. IP Gateway dipasang pada masing-masing sub-interface.

3.3 DHCP & DNS

1. DHCP: Layanan DHCP dikonfigurasi terpusat (bisa di Router atau Server). Setiap *Pool* memberikan IP, Subnet Mask, Default Gateway, dan DNS Server (192.168.40.10) .
2. DNS Internal: Server DNS di VLAN 40 dikonfigurasi dengan *Zone* pt-digitalmaju.local.
 - Record A: server -> 192.168.40.10

- Record A: www -> 192.168.40.20

3.4 NAT & Dual ISP Failover

Implementasi koneksi internet redundan :

1. NAT Overload (PAT): Mengizinkan IP Private (192.168.0.0/16) diterjemahkan ke IP Public saat keluar melalui interface Gi0/0 (ISP1) atau Se0/0/0 (ISP2).
2. Floating Static Route (Failover):
 - Route Utama: ip route 0.0.0.0 0.0.0.0 203.0.113.1 1 (Distance 1 - Prioritas Tinggi).
 - Route Backup: ip route 0.0.0.0 0.0.0.0 198.51.100.1 10 (Distance 10 - Prioritas Rendah).
 - *Mekanisme*: Router akan memilih jalur ISP1 karena *distance*-nya lebih kecil. Jika link ISP1 mati, router otomatis memindahkan trafik ke ISP2.

4. Keamanan Jaringan

4.1 Matriks ACL (Access Control List)

Kebijakan keamanan antar departemen diterapkan menggunakan *Extended ACL* pada Router Gateway :

Source	Destination	Protocol	Action	Alasan Keamanan
192.168.10.0 (HRD)	192.168.20.0 (Keuangan)	IP	DENY	Data sensitif antar dept. harus terpisah.
192.168.20.0 (Keuangan)	192.168.10.0 (HRD)	IP	DENY	Data sensitif antar dept. harus terpisah
192.168.60.0 (IoT)	192.168.40.0 (Server)	IP	DENY	Mencegah IoT yang rentan menyerang server.
192.168.30.0 (IT)	Any	IP	PERMIT	Admin butuh akses penuh untuk maintenance.
Any	Any	IP	PERMIT	Mengizinkan akses internet.

4.2 Port Security

Kebijakan keamanan fisik pada port Switch :

- Target: Port access (PC HRD, Keuangan, Staff).
- Maksimum MAC: 1 (Satu port hanya untuk satu perangkat terdaftar).
- Violation Mode: Shutdown (Port akan mati total jika terdeteksi MAC Address asing/baru).
- Respon Insiden: Jika port mati (*err-disable*), admin IT harus memverifikasi perangkat, lalu mereset port secara manual (shutdown lalu no shutdown).

5. Wireless & IoT

SSID	Security	Password
Corporate	WPA2-PSK	corporatestaff
BYOD/Guest	WPA2-PSK	guestwifi
IoT	WPA2-PSK	iotnetwork

6. Testing & Verifikasi

1. Testing Konektivitas Dasar

- Uji ping dari PC HRD ke Gateway HRD :

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- Ping dari PC HRD ke PC Keuangan (gagal ACL):

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- Uji ping PC IT ke PC HRD dan Keuangan (bisa):

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- Uji ping PC HRD ke Server:

```
C:\>ping 192.168.100.11

Pinging 192.168.100.11 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.100.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- Ping ke 8.8.8.8:

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time=3ms TTL=254

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

- Uji ping PC Branch ke Gateway Branch:

```

C:\>ping 192.168.200.1

Pinging 192.168.200.1 with 32 bytes of data:

Reply from 192.168.200.1: bytes=32 time<1ms TTL=255
Reply from 192.168.200.1: bytes=32 time<1ms TTL=255
Reply from 192.168.200.1: bytes=32 time<1ms TTL=255
Reply from 192.168.200.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.200.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

- Uji ping PC Branch ke Server:

```

C:\>ping 192.168.100.11

Pinging 192.168.100.11 with 32 bytes of data:

Reply from 192.168.200.1: Destination host unreachable.
Reply from 192.168.200.1: Destination host unreachable.
Reply from 192.168.200.1: Destination host unreachable.
Reply from 192.168.200.1: Destination host unreachable.

Ping statistics for 192.168.100.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

2. Testing Failover ISP

- Ping ke ISP 1:

```

C:\>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:

Reply from 10.10.10.1: bytes=32 time<1ms TTL=254
Reply from 10.10.10.1: bytes=32 time<1ms TTL=254
Reply from 10.10.10.1: bytes=32 time<1ms TTL=254
Reply from 10.10.10.1: bytes=32 time<1ms TTL=254

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

- Ping ke ISP 2 setelah ISP 1 dimatikan:

```
C:\>ping 20.20.20.1

Pinging 20.20.20.1 with 32 bytes of data:

Reply from 20.20.20.1: bytes=32 time<1ms TTL=254
Reply from 20.20.20.1: bytes=32 time<1ms TTL=254
Reply from 20.20.20.1: bytes=32 time=1ms TTL=254
Reply from 20.20.20.1: bytes=32 time<1ms TTL=254

Ping statistics for 20.20.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

•

- Show IP Route:

```

Gateway-HQ>enable
Gateway-HQ#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 20.20.20.1 to network 0.0.0.0

    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.20.20.0/30 is directly connected, GigabitEthernet0/1
L       20.20.20.2/32 is directly connected, GigabitEthernet0/1
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.1.0/30 is directly connected, Serial0/0/0
L       172.16.1.1/32 is directly connected, Serial0/0/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/2.1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/2.1
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/2.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/2.10
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/2.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/2.20
    192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/2.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/2.30
    192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/2.40
L       192.168.40.1/32 is directly connected, GigabitEthernet0/2.40
    192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.100.0/24 is directly connected, GigabitEthernet0/2.100
L       192.168.100.1/32 is directly connected, GigabitEthernet0/2.100
S*    0.0.0.0/0 [2/0] via 20.20.20.1

```

3. Testing Layanan Jaringan

- Cek IP klien (IP, gateway, DNS, domain) untuk memastikan DHCP berfungsi.

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: pt-digitalmaju.local
    Link-local IPv6 Address . . . . .: FE80::201:C7FF:FEB0:2889
    IPv6 Address . . . . . : ::
    IPv4 Address. . . . . : 192.168.10.10
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : ::
                                192.168.10.1
```

PC HRD

4. Testing Keamanan

- Uji ACL dari PC HRD ke Keuangan (harus gagal):

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- Traceroute ke Internet:

```
C:\>tracert 8.8.8.8

Tracing route to 8.8.8.8 over a maximum of 30 hops:

  0  0 ms    0 ms    1 ms    192.168.10.1
  1  *        0 ms    0 ms    8.8.8.8

Trace complete.
```